

Yutong Huang

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EDUCATION

University of California, San Diego, La Jolla, CA

Sep 2020 – Current

Pursuing PhD Degree in Computer Science (WukLab, advised by Prof. Yiyang Zhang)

Master's degree in Computer Science (2023)

Purdue University, West Lafayette, IN

Aug. 2014 – Dec. 2017

Bachelor of Science in Computer Engineering

Dean's List & Semester Honors (6 semesters, Spring 2015 – Fall 2017)

SKILLS

Coding Languages: C (User level and kernel level), C++, Python, Bash, CUDA, Verilog, Java

Frameworks/Tools: Linux, Pytorch, Nvidia Nsight, eBPF, LLVM, DynamoRIO, RDMA, Vivado

RESEARCH EXPERIENCE

Operating System for LLM Agent *Project Leader*

Oct 2025 – May 2026

- Extended CRIU (a container checkpointing tool) to instantly branch live GUI containers for computer-use agents by decoupling branch creation from asynchronous checkpoint serialization.
- Modified the Linux kernel for copy-on-write memory deduplication across container boundaries and page-cache deduplication across filesystems snapshots, eliminating redundant in-memory copies of the same files.
- Evaluated on OSWorld, showing up to **2.3x** faster end-to-end agent task latency vs. container-native checkpoint and **3.7x** vs. VM snapshots; memory footprint drops from 14 GB to 9 GB at 16 concurrent branches.

Multi-Tier SLO LLM Serving System *Project Contributor*

Apr 2025 – Dec 2025

- Designed a discrete-event simulator to evaluate scheduling algorithms for reasoning tokens to optimize scheduling strategies to meet strict multi-tier SLOs.
- Analyzed prompt vs. reasoning length correlations across multiple models and datasets to characterize token generation patterns.
- Validated an asynchronous, double-buffered KV-cache migration system, utilizing CUDA debugging tools to identify race conditions and ensure memory safety during high-throughput transfers.

Linux based Far Memory Prefetch with ML *Project Leader*

Jan. 2023 – Sep 2025

- Proposed ideas of separating program context and runtime memory system context to greatly increase the predictability of memory access pattern.
- Prototyped an LLVM compiler-based proof of concept, achieving 30%–70% better performance than state of the art solutions.
- Implemented the Linux kernel by modifying the kernel swap path, swap cache memory management system, and kernel swap eviction policies.
- Built an RDMA based kernel module as the far memory backend.

Memory Efficient DNN Training *Project Leader* May. 2021 – Dec. 2022

- Identified the memory bottleneck of a DNN training jobs with FLOPS calculation and Nsight profiling.
- Proposed an SGD variant to reduce memory footprint where backward pass activation data is approximated rather than stored in memory.
- Integrated this SGD with pipeline parallelism to increase GPU utilization.

Hardware-Software co-design system *Project Contributor* Sep. 2020 – Apr. 2021

- Designed user-friendly FPGA interface, enabling programmers with less hardware knowledge writing FPGA program under this memory system.
- Implemented a simple pointer tracing example on FPGA memory system, reaching 100 Gbps line rate.

LegoOS Distributed System (Best Paper) *Major Contributor* Jun. 2017 – Sep. 2018

- Achieved a single user application running on distributed hardware without modification of the application by designing and implementing distributed user space virtual memory management using C.
- Reduced distributed system failure rate and increasing application memory access parallelism by designing memory replication using an asynchronous mechanism to hide performance overhead.
- Resolved physical processors and memories resource allocation issue by designing a resource management system as a Linux kernel module.

PUBLICATIONS

An Early Exploration of Deep-Learning-Driven Prefetching for Far Memory

Yutong Huang, Zhiyuan Guo and Yiyang Zhang (NeurIPS 2025, MLsys Workshop)

LegoOS: A Disseminated, Distributed OS for Hardware Resource Disaggregation (Best Paper Award)

Yizhou Shan, Yutong Huang, Yilun Chen and Yiyang Zhang (OSDI '18)

Clio: a hardware-software co-designed disaggregated memory system

Zhiyuan Guo, Yizhou Shan, Xuhao Luo, Yutong Huang, Yiyang Zhang (ASPLOS '22)

Learned: Operating Systems

Yiyang Zhang, Yutong Huang (ACM SIGOPS Operating Systems Review)

WORK EXPERIENCE

Microsoft Research (Mentored by Sameh Elnikety) *Intern Researcher* May. 2019 – Aug. 2019

- Analyzed Latency behavior of heterogeneous virtual machine over-subscriptions in cloud environment (standard VMs and credit based VMs).
- Designed micro-benchmark for evaluating the behavior of oversubscription of IO intensive workload and CPU intensive workload.
- Evaluated disk IO latency of Hype-V hypervisors under oversubscription and compared results with KVM.

AWARDS

Jacob School of Engineering Fellowship Sep 2020

Jay Lepreau Best Paper Award at OSDI '18 Oct 2018

USENIX Student Grant for OSDI '18 Sep 2018